

AllmeD_v36 (Allme_Windows)

Comprehensive Technical Manual

Master Desktop Widget combining ASwitch + captureME + Telemetry Suite

Always-on-top, translucent circular widget with window/tab switching, screen capture, video recording & system telemetry.

1. Introduction & Uses

Allme is designed to be an omnipresent, always-on-top desktop widget. It consolidates essential system utilities into a minimal, translucent circular interface. The primary uses include rapid window and tab switching, capturing screenshots, recording the screen with audio, and monitoring system telemetry (CPU, GPU, RAM, Network). By running in the background and overlaying the desktop, it provides instant access to these tools without disrupting the user's workflow.

2. Libraries Used

- PyQt5.QtCore (Qt, QPoint, QPointF, QRectF, QTimer, pyqtSignal, QEvent)
- PyQt5.QtGui (QPainter, QColor, QPen, QBrush, QIcon, QPixmap, QCursor, QRadialGradient, QLinearGradient)
- PyQt5.QtWidgets (QApplication, QWidget, QVBoxLayout, QHBoxLayout, QSystemTrayIcon, QMenu, QAction, QSlider, QWidgetAction, QLabel, QActionGroup, QDialog, QGridLayout, QPushButton, QComboBox, QCheckBox, QGroupBox)
- PyQt5.QtWinExtras (QtWin)
- base64
- collections
- ctypes
- ctypes.wintypes
- cv2
- io
- json
- math
- mss
- mss.tools
- numpy
- os
- psutil
- socket
- subprocess
- sys
- threading
- time
- urllib.parse (urlparse)

- urllib.request (urlopen)
- winreg

3. Classes and Components

Class: `_DummyWriter`

No description available.

Key Methods:

- write
- flush

Class: `PersistentMenu`

No description available.

Key Methods:

- `__init__`
- mousePressEvent

Class: `SystemPowerStatus`

No description available.

Class: `MouseHook`

No description available.

Key Methods:

- `__init__`
- install
- uninstall
- `_hook_callback`

Class: `KeyboardHook`

No description available.

Key Methods:

- `__init__`
- install
- uninstall
- `_hook_callback`

Class: `AudioMonitor`

No description available.

Key Methods:

- `__init__`
- `run`

Class: TelemetryMonitor

No description available.

Key Methods:

- `__init__`
- `record_key`
- `record_mouse`
- `get_sample`
- `run`

Class: VideoRecorder

No description available.

Key Methods:

- `__init__`
- `start_recording`
- `stop_recording`
- `_record_loop`

Class: FaviconManager

No description available.

Key Methods:

- `__init__`
- `request`
- `poll_result`
- `_worker`
- `_parse_domain`
- `_get_browser_url`

Class: DashboardTitle

No description available.

Key Methods:

- `__init__`
- `_tick`

- paintEvent

Class: DashboardWindow

No description available.

Key Methods:

- __init__

Class: AllmeButton

No description available.

Key Methods:

- __init__
- _calc_base_size
- _zone
- _tick
- paintEvent
- _paint_impl
- mousePressEvent
- mouseMoveEvent
- mouseReleaseEvent
- contextMenuEvent
- leaveEvent

Class: AllmeWidget

No description available.

Key Methods:

- __init__
- _on_key_pressed
- _enforce_always_on_top
- showEvent
- changeEvent
- _poll
- _update_animation
- take_screenshot
- toggle_recording
- quick_switch
- cycle_left
- cycle_right
- _do_cycle

- _focus_browser
- tab_switch
- tab_cycle_left
- tab_cycle_right
- toggle_lock_position
- toggle_clickthrough
- _set_clickthrough_mode
- _update_clickthrough_state
- toggle_fly_mode
- _update_fly
- _on_global_middle
- _create_sub_menu
- _setup_tray
- _lbl
- _toggle_ontop
- _toggle_base_glow
- _toggle_glow
- _toggle_breathing
- _toggle_mouse_movement
- _toggle_startup
- _set_color_hue_target
- _set_audio_effect
- _set_telemetry_source
- _set_telemetry_target
- _set_config_val
- _on_opacity
- _on_glow_opacity
- _on_glow_size
- _on_hue
- _on_breathing_speed
- _on_size
- show_dashboard
- _on_tray
- closeEvent

4. Core Functions

Function: `get_system_accent_color`

No description available.

Function: `kill_previous_instances`

Terminates any previously running instance of Allme so only the newly launched instance runs.

Function: `log_crash_report`

Writes a dual Layman + Technical crash log whenever an unhandled exception occurs.

Function: `global_exception_handler`

No description available.

Function: `load_config`

No description available.

Function: `save_config`

No description available.

Function: `set_startup`

No description available.

Function: `check_startup`

No description available.

Function: `get_window_title`

No description available.

Function: `get_window_icon`

No description available.

Function: `is_user_application`

No description available.

Function: `get_user_windows`

No description available.

Function: `get_active_window_id`

No description available.

Function: `switch_to_window`

No description available.

Function: `is_valid_window`

No description available.

Function: `_proc_name`

No description available.

Function: `is_browser`

No description available.

Function: `send_ctrl_pageup`

No description available.

Function: `send_ctrl_pagedown`

No description available.

Function: `lerp_color`

No description available.

5. Context Menu & Interaction

The widget interacts primarily via mouse events. A right-click on the main widget typically invokes a comprehensive context menu (QMenu) providing access to:

- **Dashboard & Telemetry:** View detailed system metrics.
- **Capture & Record:** Tools for taking screenshots and starting/stopping screen recordings.
- **Settings:** Configuration options such as startup behavior and visual preferences.
- **Exit:** Safely terminate the application and its background processes.